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Question 1 – Intelligent Agent (PEAS)

1. (CO1) A smart security robot patrols a school building. It can detect whether a room is safe or unsafe and can move between rooms.

Task:

(a) Identify the **PEAS** (Performance Measure, Environment, Actuators, Sensors) of the robot.
(b) State whether the environment is **fully or partially observable**, and **deterministic or stochastic**, with reasons.
Answer:

Smart Security Robot

Performance Measure (P):

- Number of unsafe rooms correctly identified
- Coverage of all rooms in the building
- Speed and efficiency of patrol
- Minimizing false alarms

Environment (E):

- School building with multiple rooms
- Safe or unsafe conditions (e.g., intruders, hazards)

Actuators (A):

- Wheels/legs for movement
- Alarm system (sound/light)
- Communication device to alert authorities

Sensors (S):

- Cameras (visual detection)
- Motion sensors
- Heat sensors
- Gas/smoke detectors

Environment Type:

- **Partially Observable:** The robot cannot see the entire building at once; it only perceives the room it is in or nearby.
- **Stochastic:** Outcomes may vary due to unpredictable factors (e.g., intruder movement, sensor noise, environmental changes).

Question 2 – Breadth-First Search (BFS)

2. (CO1) A student wants to reach the classroom from the school entrance in a **3 × 3 grid**. One cell is blocked due to construction.

Constraints:

- Movement allowed: Up, Down, Left, Right
- The blocked cell cannot be crossed
- Each move costs **1**

Task: Apply **Breadth-First Search (BFS)** to find the shortest path from **Start (S)** to **Goal (G)**. Show the order of node expansion, the final path, and the total cost.

Answer:

Scenario: 3 × 3 grid, one blocked cell, movement allowed in 4 directions, cost = 1 per move.

BFS Characteristics:

Explores level by level.

Guarantees shortest path in terms of steps.

Steps:

- Start at **S** (Entrance).
- Expand neighbors in order: Up, Down, Left, Right.
- Skip blocked cell.
- Continue until Goal (G) is reached.

Order of Node Expansion (example): S → (neighbors) → next layer → … → G

Final Path (example layout): Suppose blocked cell is (2,2). Path could be:

S → (1,2) → (1,3) → (2,3) → G

Total Cost: Number of moves = 4 → **Cost = 4**

Question 3 – Depth-First Search (DFS)

3. (CO1) A robot is placed in a **3 × 4 maze** and must reach the goal cell (**G**) from the start cell (**S**). Some cells are blocked by walls.

Constraints:

Movement allowed: Up, Down, Left, Right
The robot cannot move through blocked cells
Each move costs **1**

Task: Apply **Depth-First Search (DFS)** to find a path from **S** to **G**. Show the order in which cells are expanded (using a stack), state the final path found, and explain why **DFS may not always find the shortest path**.

Answer:

Scenario: 3 × 4 maze, some blocked cells, movement allowed in 4 directions.

DFS Characteristics:

- Uses a **stack** (LIFO)
- Explores one branch deeply before backtracking
- May find a path, but not guaranteed shortest

Steps:

- Push **S** onto stack.
- Pop top, expand neighbors.
- Push valid moves (skip blocked).

- Continue until **G** is reached or stack empty.

Order of Expansion (example): S → (1,2) → (1,3) → (2,3) → backtrack if blocked → (3,3) → G

Final Path (example): S → (1,2) → (2,2) → (3,2) → G